MIL-T-11591D

22 August 1969

SUPERSEDING
MIL-T-11591C
18 September 1964

MILITARY SPECIFICATION

TAMPER, ARMATURE COIL WINDING

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

* 1.1 Scope.- This specification covers armature coil winding tampers (see 6.1).

* 1.2 Classification.- The tampers shall be of the following sizes, as specified (see 6.2).

Size 1 - 3/16 inch width by 7/8 inch length of face

Size 2 - 5/16 inch width by 1-1/4 inch length of face

Size 3 - 3/8 inch width by 2 inch length of face

2. APPLICABLE DOCUMENTS

* 2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-S-698	-	Steel, Sheet and Strip, Low Carbon
QQ-T-570	-	Tool Steel, Alloy
PPP-B-566	-	Box, Folding, Paperboard
PPP-B-601	-	Box, Wood, Cleated Plywood
PPP-B-621	-	Box, Wood, Nailed and Lock Corner
PPP-B-676	-	Box, Set-Up
PPP-T-45	-	Paper, Gummed, Paper, Reinforced and Plain, For Sealing and Securing

FSC 5120

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MILITARY

MIL-P-116 - Preservation, Methods Of
MIL-L-10547 - Liners, Case, and Sheet, Overwrap; Water Vapor Proof or Waterproof, Flexible

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

* 2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

E 18-65 - Methods of Test for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC.,

National Motor Freight Classification

(Application for copies should be addressed to American Trucking Association, Attn: Tariff Order Section, 1616 P Street, N.W., Washington, D.C. 20036.)

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee Agent, Room 202, Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

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(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

* 3.1 First article.- When specified (see 6.2), before production is commenced, a preproduction sample tamper shall be submitted to the contracting officer or his authorized representative for approval in accordance with 4.2. The approval of the preproduction sample authorizes the commencement of production, but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The preproduction sample shall be manufactured in the same facilities to be used for the manufacture of the production items.

* 3.2 Standard product.- The tampers covered by this specification shall be the regular commercial product of the manufacturer except for changes necessary to comply with the specification requirements.

* 3.3 Materials.- Materials not definitely specified shall be of the quality used by the manufacturer for tampers, provided the completed item complies with all provisions of this specification.

* 3.3.1 Tool steel.- The tool steel shall conform to class S1 of QQ-T-570.

* 3.3.2 Low carbon steel.- The low carbon steel shall conform to Steel Designations 1018 or 1020 of QQ-S-698.

* 3.4 Design and construction.- Tampers shall conform to the design and dimensions shown in figure 1. Size 1 tampers shall be fabricated of the tool steel specified in 3.3.1. Size 2 and size 3 tampers shall be fabricated of the low carbon steels specified in 3.3.2 and shall be carborized and pack hardened to a depth of .015 ± .003 inch. When tested as specified in 4.4.1, the size 1, 2 and 3 tampers shall have a hardness of Rockwell C48 to 55 inclusive, or equivalent.

* 3.5 Marking for identification.- The supplier shall permanently and legibly indent his name or trademark of such known characters easily identifiable on each tamper.

* 3.6 Finish.- Tampers shall be plated in accordance with the manufacturer's current commercial practice.

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* 3.7 Workmanship.- The finished tamper shall be free of burrs, pits, cracks, rough or sharp edges, or warped surfaces. There shall be no voids in plating.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 First article inspection.- When a preproduction sample is required, it shall be examined for the defects in table I, specified dimensions, and tested as specified in 4.4.1.

* 4.3 Inspection.- Sampling for inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated hereinafter.

* 4.3.1 Component and material inspection.- In accordance with 4.1, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.3.2 End product inspection.- A lot shall consist of all tampers of one size offered for inspection at one time. The sample unit for this inspection shall be one complete tamper.

4.3.2.1 Visual examination.- Examination shall be made of the tampers for defects classified in table I. The inspection level shall be level II with an acceptable quality level (AQL) of 2.5 for major defects and 4.0 for total defects, expressed in terms of defects per hundred units.

TABLE I.- Visual examination of defects

Examine	Defect	Classification Major	Minor
Finish	Not plated	X	
	Void in plating		X

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TABLE I.- Visual examination of defects (cont'd)

Examine	Defect	Classification Major	Minor
Workmanship	Burr, pit, crack, rough or sharp edge, or warped surface	X	
Marking	Missing, not legible		X

* 4.3.2.2 Dimensional examination.- Inspection shall be made of the end item for compliance with dimensions and tolerances specified. Any deviation from specified requirements shall constitute a defect. The inspection level shall be S-2 and the AQL shall be 4.0, expressed in terms of defects per hundred units.

* 4.3.2.3 Testing of the end item.- The end item shall be tested as specified in 4.4.1. The inspection level shall be S-3 and the AQL shall be 2.5, expressed in terms of defects per hundred units.

4.3.3 Examination of preparation for delivery.- An examination shall be made to determine that preservation, packaging, packing and marking as required by section 5 of this specification are complied with. Defects shall be as indicated in table II. The sample unit shall be one shipping container fully prepared for delivery. The lot size shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 and the AQL shall be 4.0, expressed in terms of defects per hundred units.

TABLE II.- Examination of preparation for delivery

Examination	Defect
Markings, exterior and interior	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Preservation	Preservative improperly applied or missing.
Workmanship	Interior boxes bulging or distorted or inadequately closed.
Weight	Weight of contents per container is more than specified.

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* 4.3.3.1 Examination of shipping containers.- Examination for defects in closure and strapping of shipping containers shall be in accordance with the appendix of the applicable container specification.

4.4 Test.-

* 4.4.1 Hardness.- The tamper shall be tested for hardness in accordance with ASTM E 18 to determine compliance with 3.4. Two readings shall be taken and the average of the readings shall be the hardness of the sample providing both readings are within specified range. Failure of this test shall constitute a defect.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.- Preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.-

* 5.1.1.1 Preservation.- Each tamper shall be cleaned process C-1, thoroughly dried, and uncoated surfaces coated with type P-1 preservative in accordance with MIL-P-116.

* 5.1.1.2 Packaging.- Ten tampers, of one size only, preserved as specified in 5.1.1.1, shall be packaged in a snug-fitting folding paperboard box conforming to variety 1, style III or VIII, type B, class a of PPP-B-566; or in a set-up paperboard box conforming to type I, variety 1, class A, style 3 or 4 of PPP-B-676. Box closure shall be secured with 2 inch minimum width gummed paper tape conforming to type III, grade B of PPP-T-45.

* 5.1.2 Level C.- Tampers shall be preserved and packaged to afford adequate protection against corrosion, deterioration and damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets these requirements.

5.2 Packing.- Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A.- Tampers, of one size only, preserved and packaged as specified in 5.1, shall be packed in a snug fitting shipping container conforming to class 2, style 2 or 4 of PPP-B-621, or overseas type of PPP-B-601. Each shipping container shall be provided with a type I or II, grade C case liner conforming to MIL-L-10547. Each shipping container shall be closed and strapped in accordance with the appendix of the applicable container specification. The weight of contents of each shipping container shall not exceed 150 pounds.

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5.2.2 Level B.- Tampers, of one size only, preserved and packaged as specified in 5.1, shall be packed in a snug fitting shipping container conforming to class 1, style 2 or 4 of PPP-B-621, or domestic type, style A or B of PPP-B-601. The weight of contents of each shipping container shall not exceed 150 pounds.

5.2.3 Level C.- Tampers, preserved and packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Rules, as applicable.

5.3 Marking.- In addition to any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.- Tampers are used to insert the coils of wire in the armature slots when motors and generators are rewound.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size tamper required (see 1.2).
(c) First article.- When a preproduction sample is required (see 3.1).
(d) Selection of applicable levels of preservation, packaging and packing (see 5.1 and 5.2).

* 6.3 The margins of this specification have been marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Air Force - 84

Review activity:

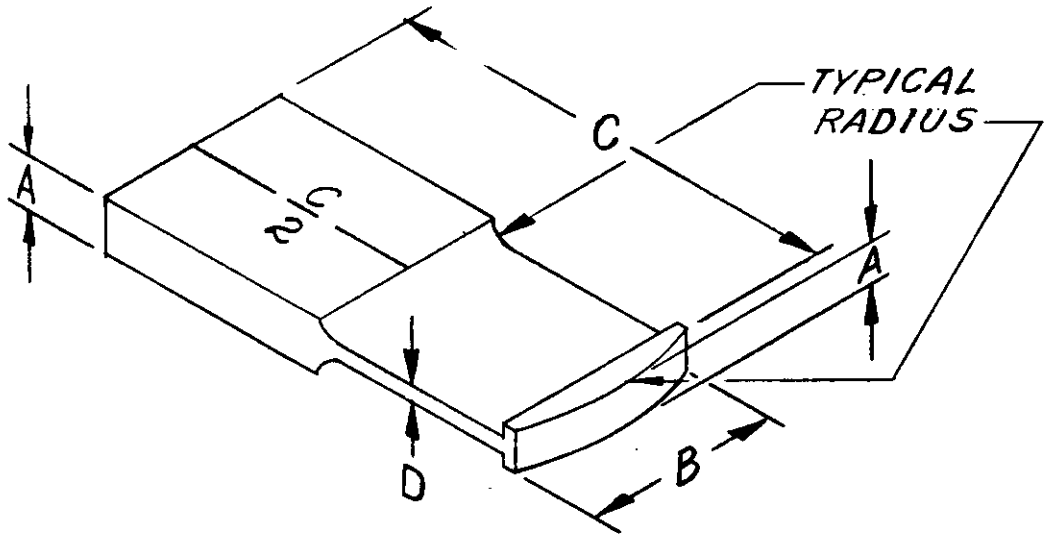
Air Force - 84

Preparing activity:

Army - GL

Project No. 5120-0672

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SIZE	DIMENSION			
	WIDTH OF FACE INCHES ±1/32 A	LENGTH OF FACE INCHES B	OVERALL LENGTH INCHES ±1/2 C	THICKNESS OF SHANK INCHES ±1/64 D
1	3/16	7/8 ± 1/8	3	1/16
2	5/16	1 1/4 ± 1/8	3	1/16
3	3/8	2 ± 1/4	3	3/32

FIGURE 1, TAMPER, ARMATURE COIL WINDING

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
Tamper, Armature Coil Winding MIL-T-11591D

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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